

# ARC FLASH FIELD HANDBOOK

Electrical Safety Reference Guide



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*Quick reference page map for field use.*

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## THANKS TO

This handbook is designed to be a quick reference and awareness for workers out in the field. This handbook is for the Walsh Group field teams, trade partners, safety professionals, QC personnel, Energy Marshals, foremen, supervisors, and craft workers who plan to work around energized equipment.

**Special thanks to:**

**The safest project is built one verified step at a time.**

# DISCLAIMER

## FIELD REFERENCE ONLY

This handbook is a field reference guide. It does not replace applicable OSHA requirements, NFPA 70E, the National Electrical Code, project specifications, employer safety procedures, energized work permits, lockout/tagout procedures, manufacturer instructions, engineered documents, or directions from the authority having jurisdiction.

This document is not an authorization to perform energized electrical work. Electrical work must be performed only by qualified persons, under approved procedures, with documented hazard/risk assessment, energy isolation, PPE, rescue planning, and supervision required by the project and applicable law.

**Where this handbook conflicts with law, contract documents, owner requirements, manufacturer instructions, or site-specific safety rules, the more protective requirement controls. Stop work and escalate any uncertainty.**

**When in doubt: STOP. VERIFY. ESCALATE.**

# FIELD SAFETY HANDBOOK

FOR TRADES & FIELD PERSONNEL

***“You don’t get a second chance with electricity.”***

## WHAT IS AN ARC FLASH?

A sudden, violent release of electrical energy through the air when electricity jumps between conductors or to ground.

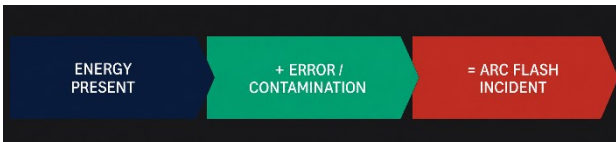


### **ARC FLASH CREATES:**

- Temps up to 35,000°F — hotter than the sun
- Intense blinding light
- Pressure blast — throws people across rooms
- Molten metal and shrapnel

## CONSEQUENCES IN THE FIELD

- Severe burns and fatalities
- Hearing loss and permanent vision damage
- Falls from blast force
- Equipment destruction and job shutdowns
- Legal exposure and project liability

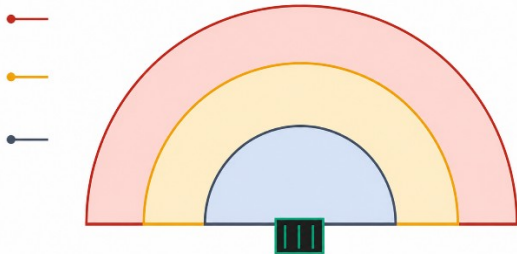


**Energy Present + Error = Arc Flash Incident**  
Remove the energy (LOTO) or the hazard (FOD).

## ELECTRICAL BOUNDARIES

Boundaries define where risk increases as you approach energized equipment. PPE requirements increase at each zone.

ELECTRICAL APPROACH BOUNDARIES



**ARC FLASH**  
PPE required

**RESTRICTED**  
Qualified only

**LIMITED**  
Increased risk



**Cross a boundary without PPE & authorization — STOP**

## CORE FIELD PRINCIPLES

### 1. Assume It's Energized

- Never trust labels or someone else's word.
- Always verify before you touch anything.

### 2. Control the Energy — LOTO

- Lock it out • Tag it out •
- Test it
- No shortcuts — ever.

### 3. Keep Equipment Clean

- Debris inside gear is a direct arc flash trigger.
- Cleanliness = life safety.



#### **4. Respect Boundaries**

- Limited • Restricted • Arc Flash Boundary
- Cross them = accept the risk.

#### **5. Read Arc Flash Labels**

- Every panel: boundary distance, incident energy, PPE, voltage.
- Don't understand it? STOP and ask.

#### **6. PPE Is Your Last Line**

- Arc-rated clothing, face shield, gloves.
- Does NOT make unsafe work safe.
- Some equipment exceeds safe PPE limits.

## 7. Follow the Process

- Planning, permits, inspections are not paperwork.
- They are what keeps people alive.



HELMET

FACE  
SHLD

GLOVES



AR SUIT



BOOTS



### **STOP WORK IF:**

- Unsure if equipment is de-energized
- Debris or contamination inside gear
- Steps were skipped or rushed
- Pressured to work energized without controls

## SECTION 1 — QC / FOD

Arc flash is often triggered by simple, preventable issues. Even a single washer can create a fault path. FOD control is life safety, not housekeeping.

**AWARENESS**  
General housekeeping

**CONTROL**  
Tool tracking, no cutting

**CRITICAL**  
Zero tolerance, sealed

### 1. Zero Tolerance Means Zero

- No debris in or around electrical equipment.
- If you see it, you own it.

### 2. FOD Is Everyone's Responsibility

- Drywall dust, shavings, packaging all count.
- Every trade contributes to the problem or the fix.

### **3. Know Your Zones**

- Awareness — general housekeeping
- Control — tool tracking, no cutting unless approved
- Critical — zero tolerance, sealed, no grinding

### **4. Clean-As-You-Go**

- HEPA vacuum required — no dry sweeping.
- No debris migration toward equipment.

### **5. Tool Control Is a Safety System**

- Account for every tool, every time.
- Tethers required when working above gear.

**6. QC Has Final Authority**

- No closure without inspection.
- No seal without verification.

**7. Tamper Seals = Trust**

- Broken seal = full reinspection required, no exceptions.

**8. Pre-Energization = Last Defense**

- No energization without documented inspection.

**NON-NEGOTIABLE RULES**

- No cutting inside electrical rooms
- No tools or materials stored on/in gear
- No open equipment left unprotected

## SECTION 1 — FOD SCENARIOS

*Use for toolbox talks and new-hire orientation.*

### SCENARIO 1 — BASIC FOD RECOGNITION

**SITUATION:** Helper notices drywall dust and metal shavings tracking toward an open panel.

**QUESTION:** *Housekeeping or your responsibility?*

**RESPONSE:** Intervene immediately — FOD is everyone's job. Stop migration, notify crew, clean before work continues.

### SCENARIO 2 — TOOL CONTROL

**SITUATION:** Electrician finishes inside a Control Area panel. No formal tool log check.

**QUESTION:** *Is visual inspection enough for closure?*

**RESPONSE:** No. Perform formal tool check, reinspect, then proceed.

**SCENARIO 3 — QC SIGN-OFF CONFLICT**

**SITUATION:** Fine dust residue in Critical Area panel. Superintendent calls it “minimal” to maintain schedule.

**QUESTION:** *Should QC sign off?*

**RESPONSE:** No. Standard is dust-free. Fine dust causes electrical tracking. Schedule is not a justification.

**SCENARIO 4 — BROKEN TAMPER SEAL**

**SITUATION:** Switchboard found with tamper seal broken. No record of authorized access.

**QUESTION:** *Can energization proceed?*

**RESPONSE:** Absolutely not. Treat as a critical event:

- Stop work immediately
- Notify GC, QC, Energy Marshal
- Full internal FOD inspection + photos
- Re-clean, reinspect, re-seal

**SCENARIO 5 — PRE-ENERGIZATION UNDER PRESSURE**

**SITUATION:** Worker reports a fastener may have been dropped inside a cabinet. No docs. Energization in hours.

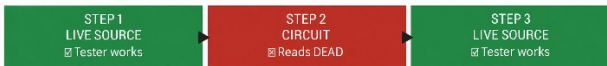
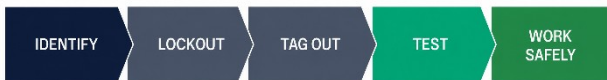
**QUESTION:** *Proceed, delay, or investigate?*

**RESPONSE:** Do not proceed. Near-miss is credible until disproven. Halt, inspect, document, reauthorize.



## SECTION 2 — ENERGY ISOLATION

Arc flash = uncontrolled electrical energy. The only real protection is controlling that energy before work begins.



**LIVE → TEST | DEAD → VERIFY | LIVE → CONFIRM  
TESTER**

### 1. De-Energize Whenever Possible

- Live work is the exception — not the plan.
- Energized work requires: infeasibility justification, risk assessment, EIP, and proper PPE.

**2. Verify Zero Energy — Live-Dead-Live**

- Test known live source → confirm tester works.
- Test circuit → verify dead.
- Re-test live source → confirm tester still works.

**3. Know Your Boundaries**

- Limited • Restricted • Arc Flash Boundary
- PPE requirement increases at each zone.

**4. PPE Is Last Line, Not First**

- Arc-rated clothing, face shield, gloves, boots.
- Some equipment exceeds safe PPE limits — escalate.

**5. Energy Isolation Permit (EIP)**

- Required for ANY energized work — no exceptions.
- Confirms justification, HRA, PPE, rescue plan, formal approval.

**6. LOTO Must Be Verified, Not Assumed**

- Confirm correct lock placement and tag ID.
- Verify all isolation points personally.

**7. Stop Work Authority Is Real**

- Anyone can stop work for unsafe conditions.
- You are expected to stop work.

## 8. Pre-Energization Is Controlled

- Gear closed and sealed. Area clear. Final inspection done.

## 9. Energy Marshal = Final Authority

- Verifies isolation and approves energization.
- Can stop unsafe work at any point.

## 10. Plan: HRA + MOP + EIP

- Level 1: Identify hazards
- Level 2: Plan — MOP, isolation, PPE
- Level 3: Execute with approvals and verification



## NON-NEGOTIABLE RULES

- No energized work without authorization
- No bypassing LOTO — under any circumstances
- No guessing isolation points
- No working alone on energized tasks

## SECTION 2 — LOTO SCENARIOS

*Use for toolbox talks and new-hire orientation.*

### SCENARIO 1 — BASIC VERIFICATION

**SITUATION:** Disconnect locked out and tagged. Crew ready to work. No one has tested the circuit.

**QUESTION:** *Safe to proceed on lockout alone?*

**RESPONSE:** No. Test for absence of voltage using Live-Dead-Live. Lockout is a control step, not proof.

### SCENARIO 2 — VERBAL CONFIRMATION ONLY

**SITUATION:** Worker: "Supervisor said it's good." No one personally verified the lockout.

**QUESTION:** *Rely on verbal confirmation?*

**RESPONSE:** No. Each worker must personally verify isolation. Stop work and confirm LOTO + zero energy.

**SCENARIO 3 — UNKNOWN ENERGY SOURCE**

**SITUATION:** Panel locked out from main. Tester shows unexpected voltage. Secondary feed suspected.

**QUESTION:** *What should the crew do?*

**RESPONSE:** Stop immediately. System is not fully isolated.

- Re-evaluate one-line diagrams
- Identify all energy sources
- Lock out all sources
- Re-test before proceeding

**SCENARIO 4 — UNAUTHORIZED ENERGIZED WORK**

**SITUATION:** Supervisor: "Tighten that connection in the energized panel just for a second." No EIP exists.

**QUESTION:** Should the worker proceed?

**RESPONSE:** No. Refuse and escalate. Energized work requires justification, HRA, PPE, and EIP approval.

**SCENARIO 5 — PRE-ENERGIZATION  
UNCERTAINTY**

**SITUATION:** "I don't remember a final verification test after removing the locks." Team pressured to energize.

**QUESTION:** *Proceed with energization?*

**RESPONSE:** No. Stop and verify.

- Reconfirm system condition
- Perform verification testing
- Ensure all clear
- Then proceed in sequence

# STOP WORK AUTHORITY

Every worker has the authority and responsibility to stop unsafe work.



## RECOGNIZE

Identify any hazard, unsafe condition, or unsafe behavior that could harm people, equipment, the environment, or the quality of the work.



## STOP

Stop the task before an incident happens. If something does not look right, pause the work and escalate the concern immediately.



## DISCUSS

Communicate the concern with supervision and the team. Work together to correct the condition, update the plan, and document how the change was implemented to help prevent it from happening again.



**Stop Work Authority may be exercised without fear of retaliation or reprisal.**



**Safety and quality depend on speaking up, stopping work when necessary, and doing the right thing.**

